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Research article

Optimizing quasi-dynamic wireless charging for urban electric buses: A two-scenario mathematical framework with grid and PV-battery systems

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ABSTRACT

This paper presents a mathematical optimization framework for the strategic placement of quasi-dynamic wireless charging (QWC) stations within road networks to address the charging needs of battery electric buses (BEBs). This study evaluates two scenarios for powering the buses. In the first scenario, a grid-connected system is considered. The optimization aims to minimize annual costs related to capital, operation, and energy losses of the electric bus fleet. This involves determining the optimal locations for QWC stations, the length of power transmitters, and the corresponding battery capacities for the BEBs. Using MATLAB-based optimization tools Casadi and Yalmip, with solvers Bonmin and Fmincon, the optimal configuration includes a 13 kWh battery capacity and a 300 m power transmitter distributed across five bus stop areas. The second scenario employs a chance-constrained optimization approach for an isolated solar photovoltaic (PV) and battery energy storage system (BESS). This system is designed to reliably meet the BEBs' energy requirements throughout the day, considering different seasonal data (winter, summer, all seasons/year-round). The optimization results for the PV and BESS capacities vary with the seasons: 394.247 kW and 2012.6 kWh using summer data, 1762.1 kW and 2738.2 kWh using winter data, and 1610.8 kW and 2741.9 kWh using year-round data. Additionally, the paper examines the impact of varying bus fleet sizes on the optimal battery size and power transmitter combination using a real-world example of the bus route between Khalifa City and Abu Dhabi Downtown in the UAE. The findings suggest that larger batteries with fewer or no charging stations are more economical for smaller fleets. Conversely, as the fleet size increases, a combination of smaller battery sizes and a greater number (and length) of QWC (power transmitters) becomes more cost-effective. This research offers significant insights into the efficient deployment of QWC stations and the integration of renewable energy and energy storage for sustainable urban electric bus networks. The proposed optimization models provide a systematic approach to designing and operating charging infrastructure, contributing to sustainable urban transportation systems. Moreover, the study highlights the influence of seasonal data on PV system sizing and costs.

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List of symbols

Sets

P_N	Set of nodes/buses in the power distribution network
P_L	Set of lines in the power distribution network
(i, j)	Set of origin and destination pairs
(g, h)	Set of scenario and hour pairs

Parameters

n	Number of buses
c_b	Cost of battery ..... USD/kWh
N_c	Number of charging stations
c_v	Variable cost of power transmitter installation ..... USD/m
c_f	Fixed cost of power transmitter installation ..... USD/m
c_b	Unit cost of battery ..... USD/kWh
r	Rate of charge of EV batteries ..... kWh/m
SOC	Battery state of charge
d_j	Amount of battery energy required to transverse two successive bus stops ..... kWh
P_j	Active power demand on bus j ..... MW
Q_j	Reactive power demand on bus j ..... MW
N_b^h	Number of buses that pass through a charging station per hour
r_{ij}	per unit resistance between bus i to j
x_{ij}	per unit reactance between bus i to j
$P_{bus,g,h}$	Power demand of wireless charging buses per hour ..... kW
η_{pv}	efficiency of the PV system
G_h	Solar irradiance per hour ..... kW/m^2
P_{max}	PV maximum power point
r_{bat}	Battery discount rate
r_{qwc}	Charging station/power transmitter discount rate
t_{bat}	Battery life span
t_{qwc}	Charging station/ power transmitter life span
λ_{losses}	Power loss cost ..... USD/MW
γ	Operating cost factor of charging stations
$P_{losses}^{initial}$	Original 13 bus system losses without the transportation system load ..... MW
X_{max}, X_{min}	Maximum and minimum possible value of wireless power track length ..... m
t	Time spent by a bus in an area ..... s/m
d	Bus energy demand between 2 consecutive stations ..... kWh

Variables

W_0	Variable representing the battery size ..... kWh
y_i	Binary variable for power transmitter placement
W_j	Battery capacity at bus station j ..... kWh
p_{ij}	Active power flow from bus i to j ..... MW
q_{ij}	Reactive power flow from bus i to j ..... MW
s_a	Binary variable for the location of EV loads in the power network
P_{pv}^{rated}	Variable representing the rated PV size
$Q_{bat,g,h}$	Residual BESS capacity at scenario g and hour h
$P_{bat,g,h}$	Charging/discharging amount at scenario g and hour h
$P_{pv,g,h}$	PV power generated at scenario g and hour h
v_{Ng}	Binary variable indicating the success or failure of each scenario for chance-constrained optimization
X_j	Power track length that would be installed at the bus stop j ..... m
v_i	Bus i voltage ..... pu

1. Introduction

Due to the world's effort towards reducing greenhouse gas emissions and combating climate change, electric vehicles are quickly replacing internal combustion engine (ICE) vehicles as a preferred alternative and many nations are currently setting action plans for phasing out ICEs soon [1–3]. However, the adoption of electric vehicles is hindered by challenges including expensive batteries, long recharging times, inadequate charging infrastructure, and power quality issues. Small electric passenger vehicles' battery expenses could amount to about one-third of the overall cost of the vehicle [4], [5]. According to a Consumer Reports survey of 3,323 people, 49% of them anticipated their electric vehicles to have a charging range of at least 300 miles [6]. A significant majority, roughly 63%, preferred a minimal 250-mile changing range. According to the research in [7] and [8], EVs could easily reach the 300-mile range with just 1% of urban roads being electrified. As a consequence, it has evolved into an unofficial yardstick for the reduction of range anxiety.

The most popular method for charging EVs, wired charging still has drawbacks, including lengthy charging periods, inconvenient wired charger use, and the requirement for larger batteries, both of which raise both the weight and price of the EV [9]. Through the use of relatively smaller battery capacities and wireless charging principles, range anxiety issues could be resolved as well as the high expense of large batteries [7], [10], [11]. As there are no cords involved, it is also much safer and more convenient than plug-in EV charging.

Three types of wireless charging exist: quasi-dynamic wireless charging (QWC), dynamic wireless charging (DWC), and static wireless charging (SWC) [7], [10–12]. Wireless charging principles make use of inductive resonant coupling-based wireless power transmission technology [13]. In static wireless charging, EVs can be charged while being parked near the charging piles. Parking lots and garages are good locations for static wireless charging because they are handy and small. The use of batteries with such large capacities is required for static wireless charging (SWC), which drives up the cost of the EV. This is due to the fact that one of the major factors driving up the cost of EVs is the cost of batteries [4], [5], [7], [14].

DWC is quickly rising in popularity as a more convenient solution to the problem of costly battery sizes. In DWC, battery electric buses (BEBs can charge while traveling on the road. The online electric vehicle (OLEV) system was created by the Korean Institute of

Science and Technology (KAIST) and is a practical DWC system on KAIST campus [13], [14]. During DWC, under the road network, charging pads and modules are built to provide electricity to moving vehicles without the need for cables. As a result, BEB batteries could be relatively smaller, which would lower the total cost of BEBs. The infrastructure and operational costs for dynamic wireless charging, however, are very expensive [7]. Another challenge of dynamic wireless charging is the efficiency of the power transmission due to the charging pads' size and coil misalignments [11].

However, in QWC, BEBs charge briefly whenever they halt along the way at bus stops or for a brief period during traffic jams. To demonstrate the potential of QWC, the automation firm ABB utilized an electric articulated bus that obtained about 400 kW charging after 15 seconds [15]. A few studies on QWC have been conducted in [7], [11], [16]. In [16], a QWC model is proposed for a grid-to-vehicle (G2V) and vehicle-to-grid (V2G) power transfer. A cost analysis of the initial investment costs for the SWC, DWC, and QWC is provided in [7]. By using QWC, it would be possible to lower the DWC charging infrastructure's high initial investment costs as well as continuing maintenance costs. The benefits made possible by QWC include convenience, reduced investment costs, increased battery capacities, and cheaper EV purchase prices. In [9], a comprehensive review of EV conducting and wireless charging infrastructure is presented. Moreover, the author presents the current and future prospects of charging infrastructure and energy resources.

The charging of BEBs also presents another problem for the electric-transport nexus if it is grid-connected. These include voltage imbalances and power losses if proper coordination is not utilized. One way of eliminating problems of grid-connected BEB charging is to supply the BEB load through an isolated power source. Renewable energy sources (RES) like solar PVs are appealing for these applications because a BEB-RES ecosystem further minimizes greenhouse gas emissions which is invaluable in combating climate change.

In this paper, the optimal siting of QWC stations in a road-electric nexus is presented. First, an optimization model is presented to optimally determine the battery size of a BEB fleet and the location of the QWC stations combination that will minimize the total annual costs. The costs include investment costs, battery costs, extra energy loss costs due to charging, and the cost of running the stations. In addition, two algorithms are used to implement the developed model, and their results are compared for the best results. Furthermore, an optimization model is developed to satisfy the energy demands of the electric buses in an isolated off-grid scenario. A PV system together with a BESS is optimally sized to achieve this feat. Additionally, the paper examines the impact of varying bus fleet sizes on the optimal battery size and power transmitter combination using a real-world example.

The main contribution of this work is to optimally site a QWC system in both a road and power network. Although other models have sited QWC systems in road networks, to the best of the authors' knowledge, a coupled road and power network for QWC systems has not been implemented in literature. Furthermore, this work also provides an approach to meet the charging demands of wireless charging EVs through a PV-BESS model taking into consideration the travel schedule of electric buses along a bus route.

2. Integrated EV, ESS and PV

Several works have been done in the area of connected EVs and PVs. These include isolated PV and EV models [17–20] as well as connected grid-PV-EV models [18], [21], [22]. These implementations have also been done to achieve several objectives. These objectives include, reviews of EV, PV and ESS technologies [18,23,22], emission reduction [21], and meeting EV energy demand.

In [17], the practicality of powering an entire power system with a PV-EV connected model is investigated. The authors sought to provide a means of eliminating traditional power plants by powering the power system with a PV and complementing the sporadic nature of PV power generation with EVs in a V2G or G2V mode. A review of the individual and combined effects of integrating solar PV systems and EVs into the power grid is presented in [23]. The authors in this work discuss how these interconnections can affect the grid's reliability, electricity quality, and economics in addition to how challenges could arise through these integrations. Furthermore, the research in [22] examines recent research on grid-connected EV-PV systems, assessing various modeling elements and approaches for estimating uncertainty in relation to these systems. The authors in [21] present a connected grid-PV-wind-EV model for emission reduction. A total number of 500,000 EVs were utilized in the model leading to an emission reduction of over 400,000 tons per year. In [19], the authors present a review of the underlying influence of the working of EV-PV systems. The technical and economic factors affecting this interaction are also presented. In [24], the authors present the performance of a PV nanogrid for control reliability and power quality, particularly for EV charging. The primary goal was to assess four distinct smart-inverter control algorithms, with an emphasis on renewable penetration, voltage profiles, net power flows, and PV curtailment, that manage battery dispatch for diverse energy management objectives. The study utilized a several measures to assess how well various BESS control modes operate, including under-voltage daily time, solar EV charging, solar penetration, and net EV demand. An economic analysis of the system is however not fully discussed. The authors in [25] develop a high-gain, fast-charging converter along with a control algorithm for efficient charging of EVs using solar power and grid electricity. The converter's performance is evaluated in various operating modes, and simulations confirm its compliance with power quality standards. The economic factors, including implementation, operating costs, and maintenance expenses, are however not covered in the study. The research in [26] focuses on the integration of photovoltaic (PV) systems in electric vehicles (EVs) to assess their impact on energy consumption, range, and charging frequency. The study, through simulations, analyzes these impacts across different vehicle types and powertrain configurations.

In [27] a two-stage robust optimization model is used to optimize solar-powered charging infrastructure for electric buses. The model also considers the degradation of charging services due to power fluctuations and extreme weather. The authors in [28] detail many facets of battery scheduling and utilization that are sustainable for electric bus networks that are connected to photovoltaic (PV) charging infrastructure. The framework employs dynamic programming, Lagrange relaxation, and a two-step approach to solve

the optimization problems related to battery replacement and fleet-depot matching. In addition, the authors in [29] explore the potential benefits of integrating Photovoltaic (PV) and Energy Storage Systems (PESS) in public transportation systems in terms of reduced operational costs and carbon emissions, as well as enhanced energy management and bus scheduling. The model utilizes a simulation-based optimization approach and incorporates a multi-source dataset including historical weather and bus operational data.

The sporadic nature of irradiance affects PV generation. Therefore, in PV-EV applications, it is incumbent to include an energy storage system (ESS) to ensure the practicality of the system. ESSs, therefore, need to be sized simultaneously with the PV systems to meet the energy demand of the EVs for these applications. The sized batteries could also smoothen the sudden differences in the output of PVs [30]. Several research works have been conducted to size ESS systems [31–36]. These works have been implemented to achieve several objectives such as the minimization of operation costs [32], [34], [35], optimal sizing of BESSs [34,37], minimization of total energy costs [31,38], and maximization of revenue [36].

An approach for PEV charging, improving the driving range of PEVs, avoiding PEV unavailability during peak demand hours, and offering a cost-effective charging solution is presented in [37]. The study analyzes the effect of battery size and load conditions on EV performance. The battery performance is assessed by utilizing the energy-to-weight ratio and the onboard energy-range factor. The authors however focused the model on an e-rickshaw which may limit the applicability to other larger EV models. The research in [38] explores the use of electric vehicles (EVs) as a temporary energy storage system (ESS) in microgrids, with additional focus on the integration with photovoltaic (PV) generation. A charging/discharging algorithm for EVs is proposed with the aim of storing excess PV energy in EV batteries or use it to supply loads when there's a shortage of PV-generated energy in order to minimize the energy imported from the grid. The model however does not address the long term economic performance and maintenance requirements of the system.

3. System modeling

3.1. Optimal siting of power transmitters in road and power network (grid connected case)

3.1.1. Objective function

The objective function of the model focuses on minimizing the total annual cost of BEB operations as well as the annual cost of losses due to BEB charging from the perspective of the transport operator. This is given by Eq. (1). The first term of the objective function describes the total annual cost of the batteries for the electric bus fleet. The second term of the objective function describes the annual construction cost concerning the power transmitters (charging stations), the third term describes the annual operating cost of the charging stations, and the last term describes the annual cost of extra power losses due to BEB charging. The annual operating cost of the charging stations is estimated as a fraction of the fixed investment and construction cost as in [39]. The parameters c_f and c_v are the fixed and variable costs of installation respectively. $P_{losses}^{initial}$ is the initial power loss before the BEB charging load is connected to the grid. The binary decision variable y_j indicates whether a charging station would be built at a bus stop j or not. Other decision variables of this optimization problem are W_0, X_j, p_{ij}, q_{ij}, and v_i^2. W_0 indicates the capacity of the battery needed for the bus whiles X_j describes the power track length that would be installed at the bus stop j, p_{ij}, q_{ij}, and v_i^2 describes the active power flow, reactive power flows and the bus voltages respectively considering the buses charge from the grid. Other parameters of the objective function are shown in Table 1.

$$\begin{aligned}\text{minimize}\Bigg[&\frac{r_{bat}(1+r_{bat})^{t_{bat}}}{(1+r_{bat})^{t_{bat}}-1}\cdot nc_b W_0+\frac{r_{qwc}(1+r_{qwc})^{t_{qwc}}}{(1+r_{qwc})^{t_{qwc}}-1}\left\{\sum_{j=1}^{N_c}c_v X_j+\sum_{j=1}^{N_c}c_f y_j\right\}\\&+\gamma\left\{\sum_{j=1}^{N_c}c_v X_j+\sum_{j=1}^{N_c}c_f y_j\right\}+8760\times\lambda_{losses}\left\{\sum_{i,j=1,j\neq i}^{P_N}r_{ij}\cdot\frac{p_{ij}+q_{ij}}{v_i^2}-P_{losses}^{initial}\right\}\Bigg]\end{aligned}\tag{1}$$

The objective function is subject to transportation network constraints and power network constraints. The two types of constraints and the algorithms implemented for solving the problem are shown next.

3.1.2. Transportation network model

The transportation network model has the following assumptions.

1. The bus stops at every bus stop
2. The energy demand between two pairs of bus stops (potential power transmitter location) is known.
3. A bus stop refers to a bus stop area.
4. Power transmitters are installed in bus station areas only.

Assumption 1 is typical of public transportation systems which usually have well-defined routes with specific bus stops. The buses along these routes are scheduled to stop at the specified bus stops irrespective of the time of day or traffic conditions. As buses tend to have specified routes, the amount of energy needed to transverse two stops could be estimated with reasonable accuracy such as in [40]. This is represented by assumption 2.

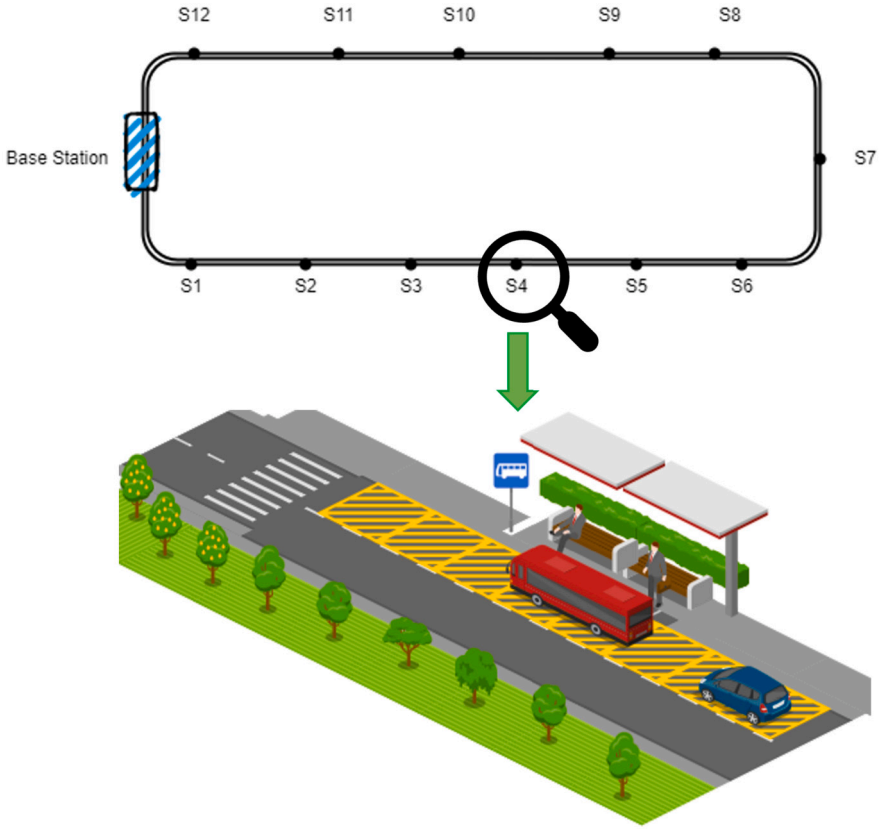


Fig. 1. Concept of bus route network.

Generally, buses tend to slow down or stop around bus stop areas. The developed QWC model takes advantage of this condition. This is because it is expected that once buses slow down or stop, time will be spent on the power transmitters to charge the electric buses. Therefore, the definition of a bus station/bus stop/bus stop area encompasses the area where the bus slows down before it stops and right after the bus starts departing. These conditions are represented by assumptions 3 and 4. Lastly, it is worth noting that the base station is equipped with a fast charger, enabling the vehicles to begin each trip fully charged, as fast chargers could charge vehicles in less than 30 minutes [41].

The transportation network constraints are adopted from [4] and we introduce a parameter t for the average amount of time spent on the power transmitters. Fig. 1 is a diagrammatic representation of the concept of the bus route. It consists of 12 bus stops which are possible charging station (power transmitter) locations, excluding the base station. The BEB can charge if the bus stop has a charging station present and the battery level is within certain limits. Fig. 1 also shows a zoom-in of a typical bus stop/charging station area where the presence of power transmitters is indicated by yellow meshes.

The state of charge (SOC) is a normalized parameter that indicates the amount of charge remaining in the battery. The maximum and minimum state of charge is given by SOC_{max} and SOC_{min} respectively. The SOC along the bus route is given by Eq. (2).

$$SOC_{min} \leq SOC \leq SOC_{max} \tag{2}$$

The maximum state of charge, SOC_{max} is considered to be proportional to the battery capacity of the buses, W_0. It is worth noting that the battery capacity W_0, the remaining onboard battery capacity at any bus stop W_j, and the amount of energy to travel between two bus stops d_j, are all measured in kWh.

Eq. (3) presents the remaining amount of energy in the battery at the first bus stop when the bus moves from the base station. d_1 is the amount of energy required to transverse the base station and the first bus station.

$$W_1 = SOC_{max} W_0 - d_1 \tag{3}$$

The amount of battery energy remaining at successive bus stations is given by the relation Eq. (4):

$$W_{j-1} = W_{j-2} + rX_{j-2} - d_{j-2}, \quad \forall j = 3, ..., N_c + 1 \tag{4}$$

where W_{j-2} describes the remaining energy at the former bus stop. The rate of charge of the battery, r, is a parameter that describes the amount of energy that is added to the battery at the bus stop area where there is an installation of a power transmitter. The

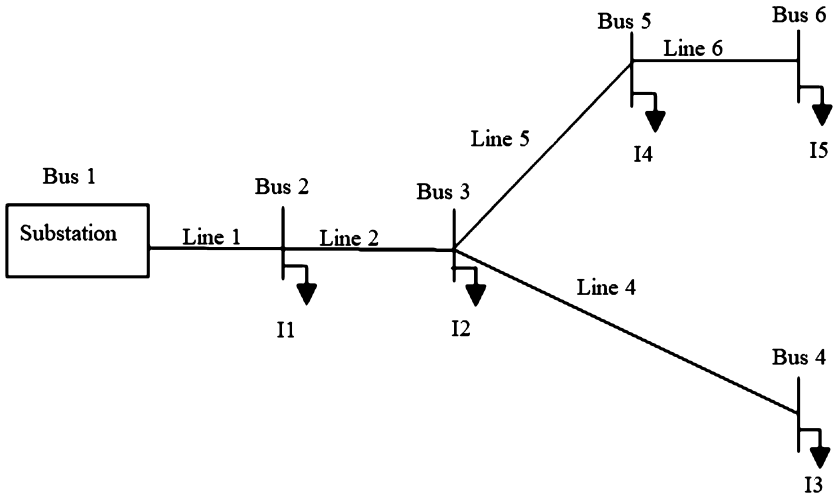


Fig. 2. Typical radial distribution network.

term rX_{j-2}, therefore, describes the total amount of energy added to the battery when the bus passes through a bus area $j-2$ of transmitter length X_{j-2}. At any point on the bus route, it is expected that the amount of energy remaining is greater than a minimum threshold. This relation is represented by Eq. (5):

$$W_j \geq SOC_{min} W_0, \quad \forall j = 1, ..., N_c \tag{5}$$

In addition, it is expected that the remaining onboard battery energy is less than the sum of the amount of charge added at a bus area, rX_j and the limit of the battery capacity, $SOC_{max} W_0$. This is given by Eq. (6).

$$W_j \leq rX_j + SOC_{max} W_0, \quad \forall j = 1, ..., N_c \tag{6}$$

The length of the power transmitters, X_j is set between two points, X_{max} and X_{min}. This ensures that the power transmitter length is realistic for a bus stop area. The limits of the power transmitters, together with the binary variable, y_j, which indicates the placement of a power transmitter at a bus station area or not, is represented by the relations below:

$$X_j \in \Re^+ \quad \forall j = 1, ..., N_c \tag{7}$$

$$y_j \in \{0, 1\} \quad \forall j = 1, ..., N_c \tag{8}$$

$$y_j X_{min} \leq X_j \leq y_j X_{max} \quad \forall j = 1, ..., N_c \tag{9}$$

By combining all the important relations for the road network described, the transportation network constraints are Eqs. (3)-(9).

3.1.3. Power network constraints

The power network could be demonstrated as a graph of nodes and links given by, $G = [P_N, P_L]$ where P_N is the set of nodes or buses for the power distribution network. The set of links, which are the distribution lines that connect various buses or nodes with each other is represented by P_L. A power distribution network is usually operated in radial topology. Therefore, because EV charging stations are on the distribution side of the power network, the radial topology is adopted for this work. A typical radial distribution network is shown in Fig. 2.

The distribution power flow (DistFlow) [42], [43], is a type of radial power flow applied mainly to distribution networks. Equations that describe the DistFlow are presented below. The ordered pair $(i, j) \in P_N$ could be described as consecutive nodes connecting the branch/link P_L. p_{ij} and q_{ij} describe the amount of active and reactive power that flow from an upstream node i to a downstream node j. p_{jn} and q_{jn} are the active and reactive power flowing from node j to a downstream node n. The active power demand at node j is represented by P_j. First, the active power balance relation is given by Eq. (10).

$$\sum_{n|(j,n)\in P_N} p_{jn} + P_j = p_{ij} - r_{ij} \frac{p_{ij}^2 + q_{ij}^2}{v_j^2}, \ \forall j \in P_N, (i, j) \in P_L \tag{10}$$

The reactive power flow balance is given by Eq. (11). The reactive power demand at a node is represented by Q_j.

$$\sum_{n|(j,n)\in P_N} q_{jn} + Q_j = q_{ij} - x_{ij} \frac{p_{ij}^2 + q_{ij}^2}{v_j^2}, \ \forall j \in P_N, (i, j) \in P_L \tag{11}$$

Then, the bus voltages could be given as Eq. (12)

$$v_i^2 - v_j^2 + 2(r_{ij}p_{ij} + x_{ij}q_{ij}) = (r_{ij} + x_{ij})\frac{p_{ij}^2 + q_{ij}^2}{v_i^2} \forall i, j \in P_N, (i, j) \in P_L \tag{12}$$

The bus voltages v_i and v_j represent the upstream and downstream voltages on a branch respectively. The parameters, r_{ij} and x_{ij} are the branch resistance and reactance of the power network respectively. A linearized form of the active and reactive power flow equations could be realized by ignoring r_{ij} and x_{ij} terms of Eqs. (10) and (11) [43], [44]. This leads to the equations:

$$\sum_{n|(j,n)\in P_N} p_{jn} + P_j = p_{ij}, \quad \forall j \in P_N, (i, j) \in P_L \tag{13}$$

$$\sum_{n|(j,n)\in P_N} q_{jn} + Q_j = q_{ij}, \quad \forall j \in P_N, (i, j) \in P_L \tag{14}$$

Also, the voltages on the buses are close to that of the reference bus for radial networks [44]. Therefore, Eq. (12) could be approximated as Eq. (15).

$$v_j = v_i + \frac{r_{ij}p_{ij} + x_{ij}q_{ij}}{v_i}, \quad \forall i, j \in P_N, (i, j) \in P_L \tag{15}$$

If the reference bus is considered as bus 1, then we can rewrite the bus voltage equation as:

$$v_j = v_i + \frac{r_{ij}p_{ij} + x_{ij}q_{ij}}{v_1}, \quad \forall i, j \in P_N, (i, j) \in P_L \tag{16}$$

To protect electric power conductors, cables, and the power network as a whole, limits are usually set for active and reactive power flows in the power network. The active and reactive power flow limit equations are represented by Eqs. (17) and (18) respectively, where the upper limit of the active and reactive power flows are $\overline{p_{ij}}$, and $\overline{q_{ij}}$, respectively and the lower limit is 0.

$$0 \le p_{ij} \le \overline{q_{ij}}, \quad \forall i, j \in P_N, (i, j) \in P_L \tag{17}$$

$$0 \le q_{ij} \le \overline{q_{ij}}, \quad \forall i, j \in P_N, (i, j) \in P_L \tag{18}$$

With the transportation network model and power network equations defined separately, there should be a relation that connects the two models. This would ensure the simultaneous placement of the QWC stations in both networks (electric-transport nexus). Such a relation could be defined by Eq. (19).

$$P_i - P_{ip} = r\sum_{j}^{N_c} n_b X_j s_a, \quad \forall i \in P_N, j \in n_c, s_a \in \{0, 1\} \tag{19}$$

Eq. (19) serves as the interface between the QWC and the power distribution network. The parameter N_c is the number of bus stops or possible charging stations (power transmitters) siting locations on the bus routes in the road network model. The parameter P_{ip} is the power demand on bus i in the original power network. The number of buses in the fleet and the rate of charging are represented by n_b and r respectively. The right-hand term is the total power needed to feed the charging stations. P_i represents the total power on bus i which is a sum of the original power in the power network and the total power for the charging stations. s_a is a binary variable indicating the placement of the power transmitter at a particular power bus or not. Eq. (19) introduces additional nonlinearities in the model.

The total number of power bus selections to supply the needed power for the transmitters is given by the constraint (20).

$$\sum_{a=1}^{N_{bus}=13} s_a = 1, \quad \forall a = 1, ..., N_{bus} \tag{20}$$

The complete optimization problem developed as a mixed integer nonlinear programming (MINLP) model is given by: minimize (1) subject to constraints (3)-(9), (13), (14), (16)-(20).

3.2. Optimal sizing of PV and BESS for the wireless charging bus fleet load

PV systems have become an increasingly important component of renewable energy adoption with a high share among all other renewable resources together with hydro and wind [45].

It could be beneficial to have an independent power source for the load of the bus fleet. This could be achieved through an isolated PV system. However, because power generation using PV is certain to be zero during the night hours, an energy storage system (ESS) is needed to complement the PV and supply power during this period. In this section, an approach for the sizing of PV and the ESS is presented for the EV bus fleet is presented. The various expressions and equations pertaining to this optimization problem are clearly developed and defined within this section. The problem is developed as a chance-constrained optimization (CCO) problem due to the uncertainties that exist in the PV model. Fig. 3 illustrates how the PV and work together in powering the power transmitters (yellow

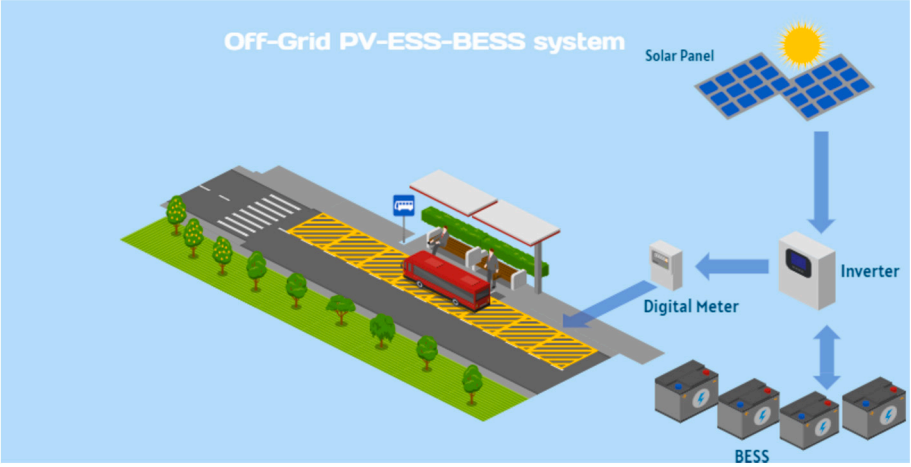


Fig. 3. Offgrid PV-BESS-BEB system.

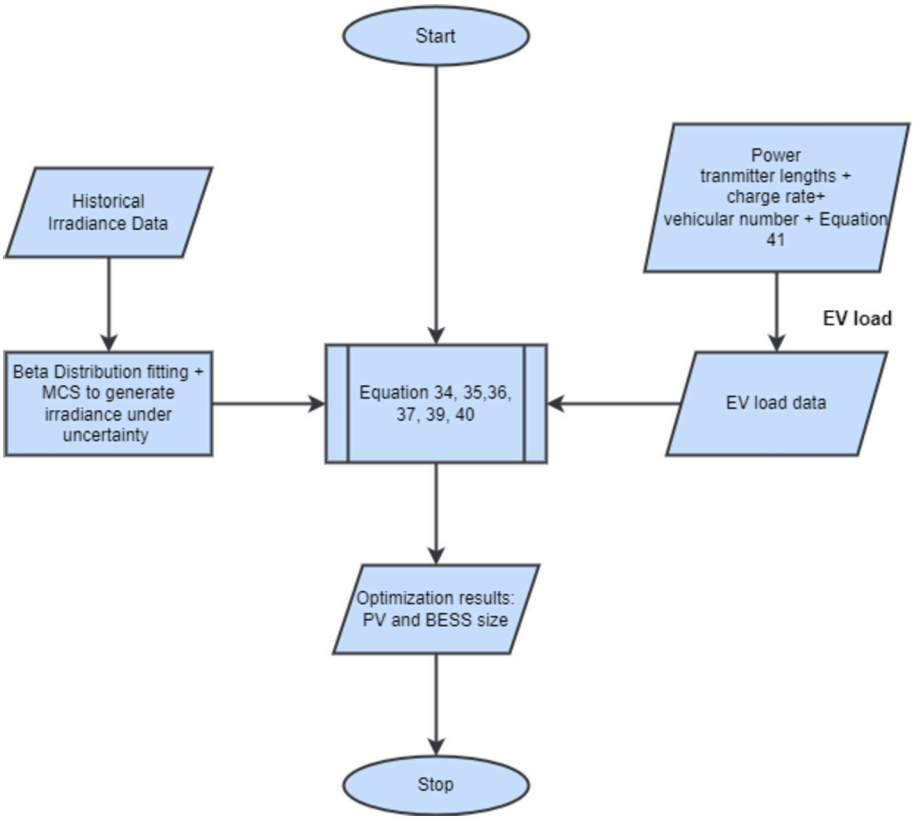


Fig. 4. Flowchart for sizing of PV and BESS.

meshes) in order to charge the BEBs. The flowchart for the sizing of the PV and the BESS to meet the bus fleet load demands is shown in Fig. 4.

3.2.1. Chance constrained optimization model

Chance-constrained optimization (CCO) provides a means of solving optimization problems with some uncertainties. We require a formulation for meaningfully limiting the values and processes described by random variables. The parameters involved in such optimization problems tend to follow a distribution, usually with tails. These distributions' random variables have such a low chance of taking on values above or below a certain threshold that it is reasonable to conclude that the likelihood of these events occurring is extremely low [46]. Generally, the idea of CCO is that soft constraints do not have to be satisfied at all costs. The constraints are

formulated such that they are satisfied to a minimum probability, given as $1-\epsilon$, where ϵ is referred to as a violation parameter. Typical values for ϵ are [0.01, 0.05, 0.1] and are chosen depending on the degree of violation permissible for the problem being solved [46].

CCO could be applied in areas of stochastic optimization such as in power engineering [47], [48], financial risk management and UAV navigation [46]. In power systems studies, CCO could be used to determine the optimal power generation under the probabilistic impact of renewable energy sources like PV and wind.

Given an initial constraint:

$$\sum_{i=1}^{n} a_i x_i \leq b \tag{21}$$

where n is the number of constraints, a and b are constants, and x is a variable, this constraint could be reformulated as a chance constraint:

$$P\Big\{\sum_{i=1}^{n} a_i x_i \leq b\Big\} \geq 1-\epsilon, \in [0,1] \tag{22}$$

Solving CCO problems could be very difficult. Given a general constraint consisting of a deterministic part x and a stochastic part ζ, the solution involves solving the integral in Eq. (23) which could be very difficult or impossible to solve.

$$\big\{[P(F(x,\zeta) \leq b] \geq 1-\epsilon\big\} = \int\limits_{\{\zeta | f(x,\zeta) \leq b\}} PDF(\zeta) d\zeta \tag{23}$$

In Eq. (23), $PDF(\zeta)$ is the probability density function of ζ and b is a constant. Due to the complexity of solving the problem in (23), approximate approaches are adopted in computing the solution. Some of these approaches include scenario-based [49], sample average approximations [50], using chance constraint surrogates [51], through Hilbert space embedding [52], back mapping [53], and polynomial chaos expansion [46].

In the scenario-based model, the worst case of the problem is considered, and it is expected that the chance constraints, given generally as $F(x,\zeta) \leq b$, be satisfied for as many scenarios as possible. With enough constraints, it is expected that the solution to the CCO problem lies in the feasible region of the original problem [46]. Further details about CCO and its solution approach can be found in [53].

Due to its relative simplicity as compared to the other models, the scenario-based approach is considered in this paper.

3.3. Development of optimization model

In this section, the step-by-step development of the PV system and BESS is presented. The various expressions and relations necessary for the models are discussed thoroughly.

3.3.1. Solar irradiance and PV model

The output power of a solar module is small. Therefore, it is important to connect many modules to form an array to increase the power output. The external characteristics of the solar cell relate to its current-voltage (I-V) dependence. This dependence is shown in Fig. 5.

In Fig. 5, V_{oc} and I_{sc} are the open circuit voltage (in volts) and the short circuit current (in ampere) respectively. V_{mp} and I_{mp} are the maximum voltage (in volts) and maximum current (in ampere) at the maximum power point P_{max}. Another quantity, the fill factor (FF) describes the quality of the solar cells through how they fit into the rectangular structure formed by V_{oc} and I_{sc} in Fig. 5. The FF is given by Eq. (24) and is practically between 0.75 and 0.9 and depends on the type of solar cells [54], [55].

$$FF = \frac{P_{max}}{I_{sc} \cdot V_{oc}} \tag{24}$$

With a solar irradiance, G, the efficiency of the solar cell could generally be given as in Eq. (25).

$$\eta_{cell} = \frac{P_{max}}{G \cdot A} \tag{25}$$

where A is the area covered by the PV array.

In this paper, the PV output equation follows the approach in [56] and is given as Eq. (26).

$$P_{pv,h} = P_{pv}^{rated} \cdot \frac{G_h}{1000} \cdot \eta_{pv} \tag{26}$$

Here, $P_{pv,h}$ represents the PV power output at a certain hour of interest, P_{pv}^{rated} is the rated capacity of the PV, G_h is the hourly irradiance, and η_{pv} is the efficiency of the PV system.

Hourly irradiance data is obtained from historical datasets. This data is fitted to a Beta distribution function. Then the Montecarlo simulation is used to randomly generate hourly irradiance data for several scenarios. This approach takes care of the uncertainties in solar irradiance which leads to uncertainties in PV output as has been applied in [57] and [58].

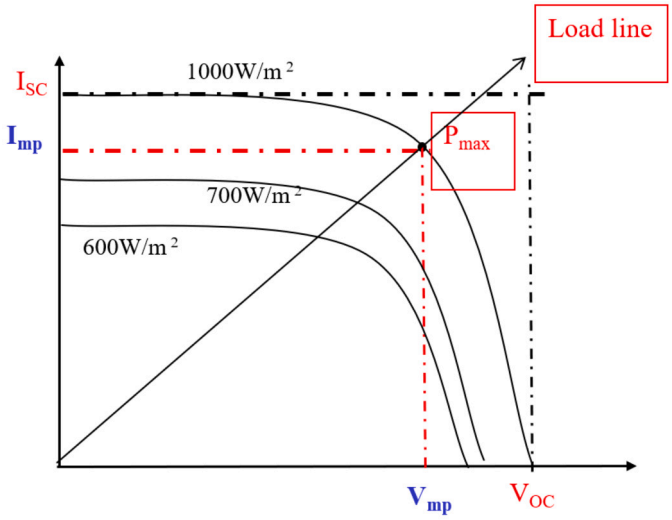


Fig. 5. I-V characteristics of a solar PV cell.

The equations applied for the Beta distribution are presented in Eqs. (27), (28), and (29) [57].

$$f_{beta}(s) = \begin{cases} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta} \cdot s^{\alpha-1} \cdot (1-s)^{\beta-1}, & \text{for } 0 \le s \le 1, \alpha, \beta \ge 0 \\ 0 & \text{otherwise,} \end{cases} \tag{27}$$

where $f_{beta}(s)$ is the Beta distribution function of s, s represents the solar irradiance in kW/m^2, and α and β are the parameters of the Beta distribution function.

To calculate the parameters α and β for the Beta distribution function, Eqs. (28) and (29) are applied.

$$\beta = (1-\mu) \cdot \left(\frac{\mu \cdot (1+\mu)}{\sigma^2} - 1\right) \tag{28}$$

$$\alpha = \frac{\mu \cdot \beta}{1-\mu} \tag{29}$$

where μ and σ are the mean and standard deviation of the irradiance respectively.

3.3.2. BESS model

The residual amount of battery energy at any scenario g during a particular hour h is given by Eq. (30) where $P^{charge}_{bat,g,h}$ and $P^{disch}_{bat,g,h}$ refer to the charged and discharged powers respectively. The parameters η^{charge}_{bat} and η^{disch}_{bat} refer to the charging and discharging efficiencies respectively. The time step given by Δt refers to the interval between consecutive time instances, taken as 1 hour in this work.

$$Q_{bat_{,g,h}} = Q_{bat_{,g,h-1}} + \left(P^{charge}_{bat,g,h}\eta^{charge}_{bat} - \frac{P^{disch}_{bat,g,h}}{\eta^{disch}_{bat}}\right)\Delta t \tag{30}$$

The relations, (31) and (32), provide the limits for the charging and discharging of the batteries. Eq. (33) prevents the charging and discharging of the batteries at the same time.

$$0 \le P^{charge}_{bat,g,h} \le P^{charge}_{bat,g,h,max} \tag{31}$$

$$0 \le P^{disch}_{bat,g,h} \le P^{disch}_{bat,g,h,max} \tag{32}$$

$$P^{charge}_{bat,g,h} \times P^{disch}_{bat,g,h} = 0 \tag{33}$$

To simplify the model, the battery self-discharging is ignored in this work. In addition, constraints (31) to (33) are reformulated as constraints (34) and (35), ignoring the charging and discharging efficiencies.

$$Q_{bat_{,g,h}} = P_{bat_{,g,h}}\Delta t + Q_{bat_{,g,h-1}} \tag{34}$$

Furthermore, constraint (35) relates to the power-to-energy capacity ratio which regulates the amount of power that could be obtained from the battery over a particular time. For example, if the energy capacity of a battery is 10 kWh and its power capacity is 2 kW, then maximum power could be produced by the battery for 5 hours. The parameter, bat_{cf} is the number of hours the battery

would last at its maximum power. The power-to-energy capacity ratio could protect the battery from deterioration and maintain its charging and discharging efficiency over a relatively longer period [59].

$$-\frac{Q_{bat}}{bat_{cf}} \leq P_{bat_{,g,h}} \leq \frac{Q_{bat}}{bat_{cf}} \tag{35}$$

In constraint (36), the minimum and maximum limits are set for the battery during charging and discharging.

$$SOC_{min}^{bat} Q_{bat} \leq Q_{bat_{,g,h}} \leq SOC_{max}^{bat} Q_{bat} \tag{36}$$

3.3.3. PV-BESS-bus load power balance

Now, to ensure that the power supplied by the PV and the power delivered or absorbed by the BESS is enough to ensure the electric bus load is catered to, a power balance relation needs to be established. This power balance is represented by constraint (37), where $P_{pv,g,h}$, $P_{bat,g,h}$, and $P_{bus,g,h}$ is the PV power generated at hour, h and scenario g, battery power (charging or discharging) at hour h and scenario g, and power demand by the electric buses at hour h and scenario g respectively.

$$P_{pv,g,h} + P_{bat,g,h} - P_{bus,g,h} \geq 0 \tag{37}$$

3.3.4. Chance constraints

The constraints (34)-(37), (39) described above take the form of a CC which can be described as:

$$F(x_{g,h}, \zeta_{g,h}) \leq M(1 - v_{Ng}) \tag{38}$$

where M is the big M parameter, taken as 10^4 in this work. As explained earlier, in CCO, it is expected that the constraints are achieved to a confidence level set by the user according to the level of relaxation of the constraints wanted. The binary variables indicating the success or failure of the constraints are represented by v_{Ng}. The level of confidence is represented by $1 - \epsilon$ with ϵ representing the degree of freedom from hard constraints. This constraint is given by Eq. (39).

$$\frac{v_1 + v_2 + v_3 + ... + v_{N_g}}{N_g} \geq 1 - \epsilon \tag{39}$$

3.3.5. Overall optimization function

The overall optimization problem could be stated as:

$$\text{Minimize } Z = cost_{pv} P_{p_v^{rated}} + cost_{BESS} Q_{bat} \tag{40}$$

subject to the constraints, (34)-(37), (39).

4. Concept exploration and simulation study

4.1. General base test case

4.1.1. Optimal siting of power transmitters in grid-connected electric-transport nexus

To test the developed model, an illustrative study is considered. First, a 12 bus stop route network is considered as shown in Fig. 1 (see Fig. 6 for a general 13 bus system). The bus stop areas are all considered possible charging station (power transmitter) locations and the following points are also considered.

1. The energy required to transverse two consecutive bus stops is known beforehand.
2. The bus stop areas cannot be longer than 60 m.
3. The buses follow an identical route plan and they all stop at all the bus stops and have similar average time spent at bus stations.

The data for the route network is adopted from [4]. The IEEE 13 bus is adopted as the radial network for DistFlow. Data for the IEEE 13 bus network can be found in [61]. The parameters of the road network are given in Table 1.

4.1.2. Optimal sizing of PV and BESS

This model aims to provide an isolated power system to serve the bus fleet load by powering the power transmitters installed at the bus stop areas. This eliminates the dependence of the bus fleet load on the grid power and the problems that could arise in the grid due to the intermittent charging of the buses.

The amount of power being generated by the PV is formulated as an $N_s \times 24$ matrix where N_s is the number of scenarios being considered, and 24 is the number of hours in the day. The solar irradiance per hour is obtained from a historical dataset obtained from [62]. To incorporate the uncertainties in the solar irradiance and thus power generation by the PV, first, the irradiance data is fitted to a Beta distribution and then the Montecarlo simulation is used to generate the hourly irradiance.

The parameters for the sizing of the PV and BESS are shown in Table 2.

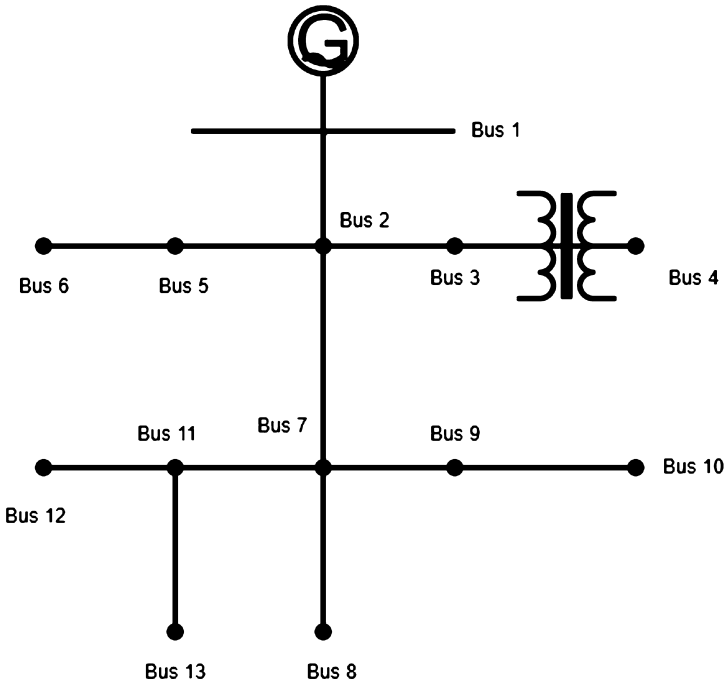


Fig. 6. General 13 bus system.

Table 1
Description of road network parameters and their values [4].

Parameter	Description	Value
c_b	Unit cost of battery	USD150/kWh [60]
c_v	Variable cost	USD30/station
c_f	Fixed cost of installation	USD 2250/station
N_c	Number of stations excluding the base	12
n	number of buses	10
t	Time spent bus in area per meter	10 s/m
r	Energy charge rate	0.1 kWh/m
d	Energy demand between two consecutive stations	4, 5, 1.67, 2, 3.33, 4, 2.33, 4, 4.33, 1.66, 4, 1, 2.66 (kWh)
X_{max}, X_{min}	Specified distance range	60 m, 0
λ_{losses}	Cost of electric power loss	USD 0.1/kWh
γ	Factor for calculating the operating cost of charging stations	0.15
r_{bat}	battery discount rate	0.08
t_{bat}	battery lifespan	10 years
r_{qwc}	charging station/power transmitter discount rate	0.08
t_{qwc}	charging station/power transmitter lifespan	15 years

Table 2
Parameters for sizing the PV and ESS.

Parameter	Description	Value
$cost_{BESS}$	Cost of BESS	USD 1000/kWh [63]
$cost_{PV}$	Cost of PV system	USD 1000/kW [64]
SOC_{max}	Maximum state of charge of BESS	0.9 [4]
SOC_{min}	Minimum state of charge of BESS	0.1 [4]
$SOC_{initial}$	Initial state of charge of BESS	0.5
η_{pv}	Efficiency of PV converter	96% [65]
ϵ	constraint failure probability	0.05 [46]

4.2. Real world test case

Next, we explore the concept applied in the base test case on a real-world bus route. As discussed in the base general model, the bus stop/areas are all considered possible charging station (power transmitter) locations. The practicality of this is seen in public transport systems such as those in Abu Dhabi and Dubai in the UAE where each bus has a specific route and specifically assigned bus

Table 3
Distance Parameters for Real Work Test Case.

From	To	Distance (km)	Energy Required (kWh)
Base station	SAN schools	2.6	0.52
SAN schools	NMC Royal	1.2	0.24
NMC Royal	National Aquarium	9.6	1.92
National Aquarium	Al Jazeera Club	14.32	2.864
Al Jazeera Club	Bint Said Mosque	3.8	0.76
Bint Said Mosque	Capital Park	4	0.8
Capital Park	ADB bus station	5.1	1.02
ADB bus station	MBK street	1.2	0.24
MBK street	Al Jazeera Club	1.7	0.34
Al Jazeera Club	National Aquarium	16	3.2
National Aquarium	Al Tashreef	9.8	1.96
Al Tashreef	NMC Royal	2	0.4
NMC Royal	Base station	2.2	0.44

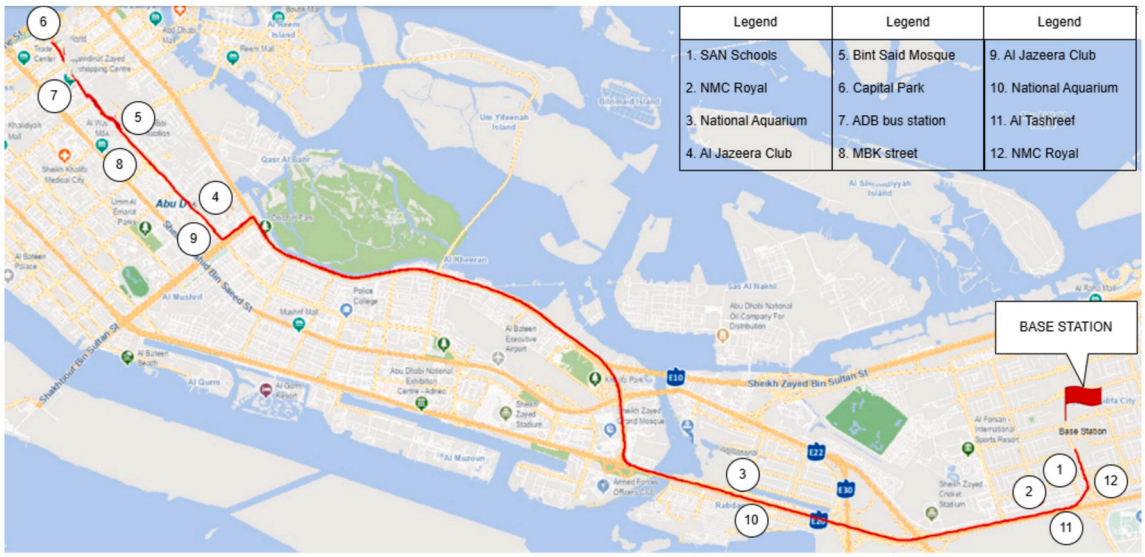


Fig. 7. Map of bus Khalifa City-Abu Dhabi Downtown bus route.

stops. These buses always stop at the designated bus stop areas, irrespective of the time of day. We employ the same assumptions listed in section 4.1.

The 'bus number 160' which operates between Khalifa City and Abu Dhabi Downtown in the UAE was selected for this case. According to [40], the average energy consumption of an EV battery is 0.2 kWh/km. This value is used to calculate the average energy between two bus stops in this case. The various bus stations, distances, and energy required to transverse consecutive bus stations are shown in Table 3 [66]. In addition, Fig. 7 [66] shows the area selected for this case. The remaining data is adopted from Table 1.

5. Results and discussion

5.1. General base test case

This section exhaustively discusses the results of the optimization models developed. First, a discussion on the placement of the power transmitters in the route network and the IEEE bus is presented. The various bus voltages from the Distflow are all graphically depicted. Then, a comparison of the results based on the two optimization approaches used in solving this model is made.

Next, the results from the isolated PV model and BESS are presented. This includes the expected size for the PV and BESS to be installed to match the load demand of the electric buses within the fleet.

The optimization problem for the optimal siting of the power transmitters and the sizing of the PV and BESS was modeled as a MINLP and mixed integer linear programming (MIP) problems respectively and solved with MATLAB. The specification for the device used for the simulation was Intel(R) Core(TM) i5-1035G1 CPU at 1.00GHz.

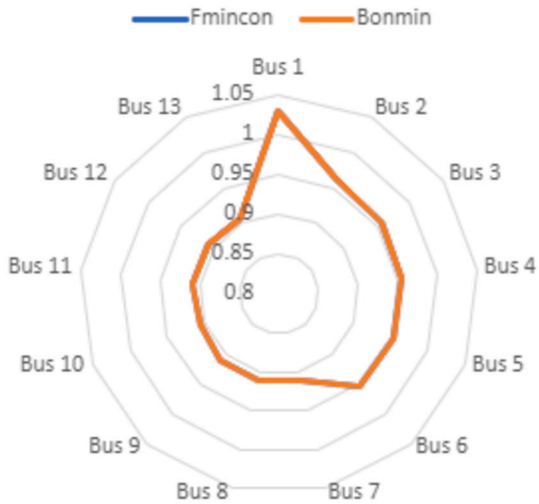


Fig. 8. Bus voltage levels.

Table 4
Optimization results of bus voltages (pu) for bus voltage levels.

Bus numbers	Fmincon	Bonmin
1	1.03	1.03
2	0.9593	0.9593
3	0.9559	0.9559
4	0.9559	0.9559
5	0.9547	0.9547
6	0.9532	0.9532
7	0.9103	0.9103
8	0.9103	0.9103
9	0.9103	0.9103
10	0.9056	0.9056
11	0.9083	0.9083
12	0.9073	0.9073
13	0.9065	0.9065
Bus Placed	Bus 1	Bus 1

5.1.1. Optimal siting of power transmitters

In relation to the IEEE 13 bus network used as shown in Fig. 6, the reference bus is set to be bus 1. The Bonmin solver through CasADI optimization toolbox is used to solve the problem. The Yalmip optimization toolbox with the fmincon solver is also used to confirm the results of the CasADI optimization.

With respect to bus voltage magnitude results, the weakest and strongest buses are bus 10 and bus 1 with a voltage magnitude of 0.9056 pu and 1.03 pu respectively. This is in line with the voltage magnitude range of 0.90 pu-1.10 pu, considered in [67–69]. A graphical representation of the various voltage levels is shown in Fig. 8. Table 4 summarizes the voltages on various buses after the power transmitter placement.

Furthermore, from Table 5, the results for the charging station (power transmitter) location are presented for the developed problem. There are five different combinations with the total length of installed power transmitters relatively close.

The length of power transmitters is inversely proportional to the battery capacity of the optimization solutions. This is due to the fact that the larger the amount of energy in a battery of a BEB, the longer the range and the less charge it would need on the road. This is confirmed by the power transmitter length results for the two solvers and their respective battery capacities. The fmincon, with the shorter power transmitter length, has a larger BEB battery size. The Bonmin solver has a relatively smaller battery size of 12.9 kWh, which corresponds to a relatively longer power transmitter length of 300 m. The fmincon solver has a relatively larger battery size of 13.325 kWh which is balanced by a relatively shorter power transmitter length of 296.6 m.

The cost of batteries makes up a huge chunk of the overall cost of EVs. The larger the battery capacity, the heavier and bulkier the BEB also is. Therefore for the application of dynamic and quasi-dynamic wireless charging principles, the aim is to find a balance where a relatively smaller battery size is used while fulfilling the extra charging demands of the BEBs through the power transmitters. Table 6 shows the objective value of the optimization problem. The Bomnin solver has a slightly better objective function value which could be attributed to the difference in the battery sizes since the battery capacity plays a higher impact on the overall investment cost of running a BEB bus fleet.

Table 5
Optimization results for charging station (transmitter) placement on road network.

Power Transmitter Length (m)		
Bus stop area	Fmincon	Bonmin
1		
2	60	60
3	60	
4		60
5		
6	60	60
7	60	
8		60
9	56.6	60
10		
11		
12		
Total length (m)	296.6	300

Table 6
Optimization results for the objective function and bus battery capacity.

Results	Fmincon	Bonmin
Object. Value (USD)	7380.70	7330.32
Bat. Cap. (kWh)	13.325	12.9

Table 7
Typical bus schedule for bus station 1.

Bus Station 1		
Bus Number	Time of first round	Time of last round
1	6:00	21:00
2	6:15	21:15
3	6:30	21:30
4	6:45	21:45
5	7:00	22:00
6	7:15	22:15
7	7:30	22:30
8	7:45	22:45
9	8:00	23:00
10	8:15	23:15

For the developed model, therefore, a 13 kWh battery capacity, is selected for the bus fleet and 5 power transmitters would be installed at 5 different bus areas with a total power transmitter length of about 300 m and a total annual cost of USD 7330.32.

5.1.2. Optimal sizing of PV and BESS

In this section, the results of the developed PV and BESS model described are presented and discussed. Due to the strong performance of the Bonmin solver, it is used in solving the developed problem.

To know the daily load profile of the bus fleet, a bus schedule is constructed with a 15-minute interval of arrival for every bus stop. Typical arrival times of the buses at bus station 1 and bus station 10 are as shown in Tables 7 and 8 respectively. It is also assumed that the first bus reaches the first bus stop at 6:00 am and the last bus arrives at the last bus stop (S12) at 23:57 pm. Therefore, each bus makes 7 round trips during the day.

From the daily bus schedule of the various bus stops, the hourly vehicular data could be computed. The hourly bus data for different charging stations are presented in Fig. 9.

The load demand is an important component of the PV sizing model. The hourly load profile of the vehicles is given in Eq. (41).

$$Hourly \quad load \quad profile, \quad kWh = rN_b^h \sum_{j=1}^{N_c} X_j \tag{41}$$

In Eq. (41), N_b^h is the number of buses that pass through a bus stop within a particular hour, r is the rate of charge, and X_j is the length of the power transmitters.

Table 8
Typical bus schedule for bus station 10.

Bus Station 10		
Bus Number	Time of first round	Time of last round
1	6:35	21:35
2	6:50	21:50
3	7:05	22:05
4	7:20	22:20
5	7:35	22:35
6	7:50	22:50
7	8:05	23:05
8	8:20	23:20
9	8:35	23:35
10	8:50	23:50

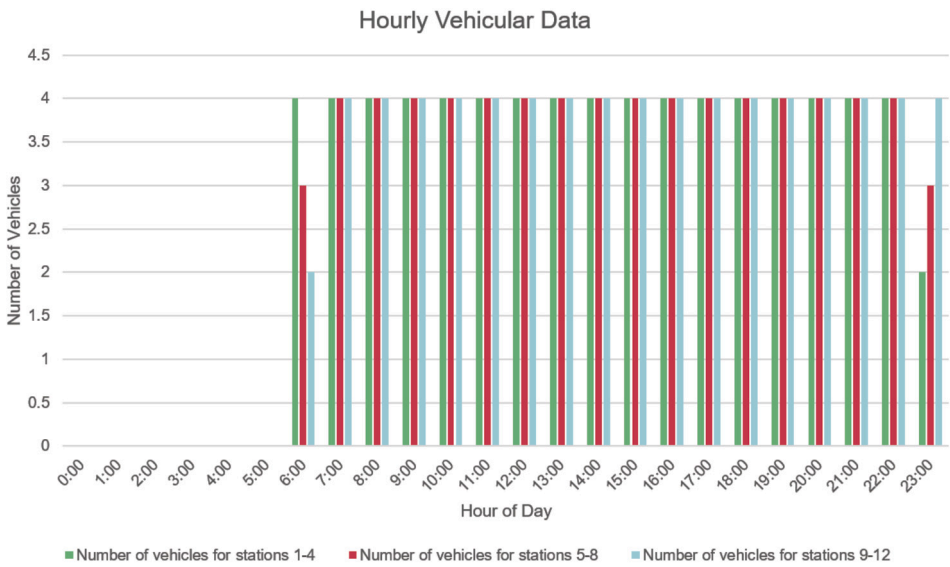


Fig. 9. Hourly vehicular data for various bus stops.

Table 9
Comparison between the developed model and other models.

Reference	Siting of station	Length of transmitters	Grid connected	PV-ESS	Comp. other solvers/model
[4]	✓	✓	x	x	x
[7]	✓	✓	x	x	x
[14]	✓	✓	x	x	x
[71]	x	x	✓	x	x
[72]	x	x	✓	✓	x
[67]	✓	✓	x	x	✓
[27]	x	x	x	✓	✓
[28]	x	x	✓	✓	✓
[29]	x	x	x	✓	✓
This work	✓	✓	✓	✓	✓

This computed profile using Eq. (41) is shown in Fig. 10. This profile indicates the amount of energy required for all buses during a one-hour period (also the power). This serves as the load input for the PV and BESS sizing model.

Solar irradiance is the key component in determining the amount of hourly output PV power.

Historical irradiance data is sourced from [70] for a location (Latitude 53.2247, Longitude -2.9641) in the UK. From the historical data, the Beta distribution fitting of the solar irradiance is determined. This is done for the winter, summer, and all seasons such that a comparison of how seasonal variations affect the PV and ESS sizing results could be made. Fig. 11, shows the Beta PDF fitting for two different hours, hour 12, and hour 13 for summer data. The results for the Montecarlo simulation, which generates the hourly irradiance for the summer period, for 100 scenarios are shown in Fig. 12.

In using the summer data, the residual battery energy is presented as Fig. 13 for all the 100 scenarios of simulation. The minimum remaining capacity of the BESS is found to be 800 kWh around the 8-hour (8am) throughout the 24-hour period. The maximum

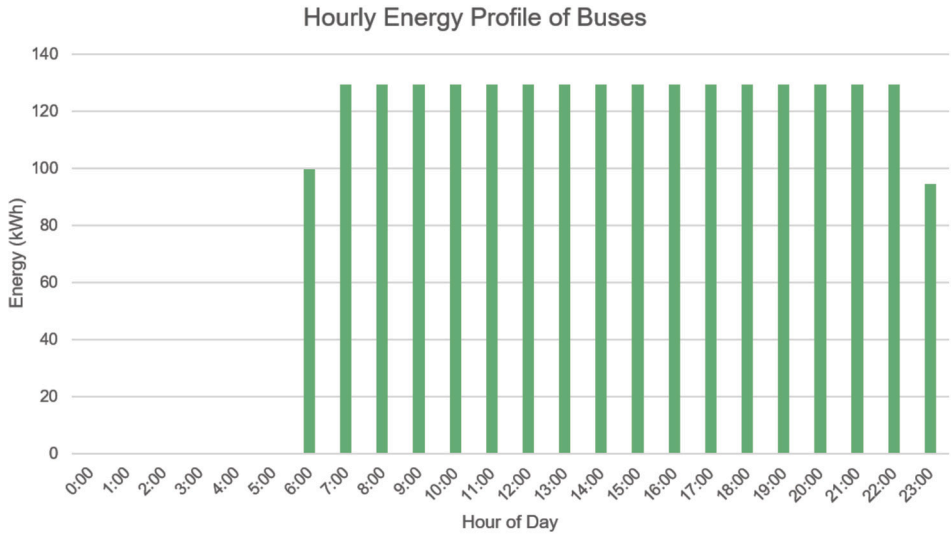
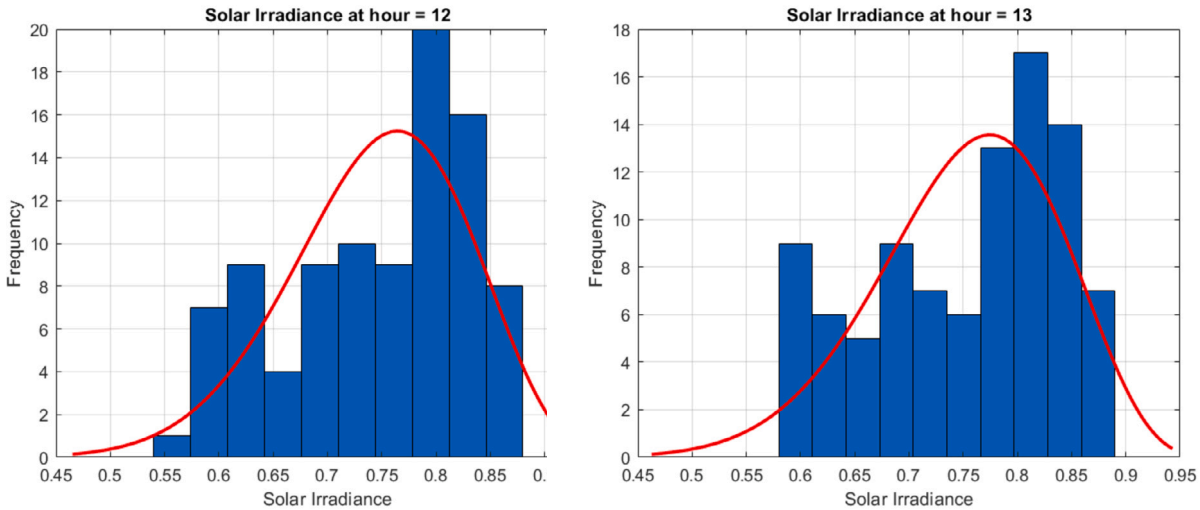


Fig. 10. Hourly load profile of charging stations.



(a) Solar irradiance histogram and Beta distribution at 12th hour (b) Solar irradiance histogram and Beta distribution at 13th hour

Fig. 11. Solar irradiance histogram and Beta distribution for summer data.

residual capacity falls within the 16-17th hours of the day. It is also important to note that, between the 0-8th hour, the BESS have their capacities falling below their initial state (1000 kWh). This is reasonable as between those hours, the irradiance levels and thus the PV output is very low or even zero, and therefore, the batteries need to discharge to support the BEB load. The batteries however have their capacities increased from 8am to about 5pm where we expect the PV to generate an output to supply the load and charge the battery. As the PV output reduces afterward, the battery supports the PV in supplying the load with the required energy needed. The process is evident through the comparison of Fig. 12 and 13. The optimized complementary PV and BESS capacities to supply the BEB loads are found to be 394.2470 kW and 2012.6 kWh respectively. It is worth noting that the maximum residual energy for the battery is about 1800 kWh which is about 90% of maximum capacity, set to protect the battery from overcharging.

Furthermore, winter irradiance data as well as year-round (all seasons) data is utilized for the PV and BESS sizing. The results of the optimization is shown in Table 11 and 12.

The winter and summer data are chosen alongside the year-round data because they provide the upper and lower boundaries for the cost of the system. From Table 10, 11, 12, it is evident that utilizing only the winter data for sizing returns the highest cost for the system. This is because winter periods have relatively the least solar irradiation as compared to other seasons. In addition, the summer period as expected had the least cost due to it being the period with the high solar irradiation. Since the all seasons (year-round) data consist of winter and summer data (which has the extreme data points of solar irradiation), it is expected that the

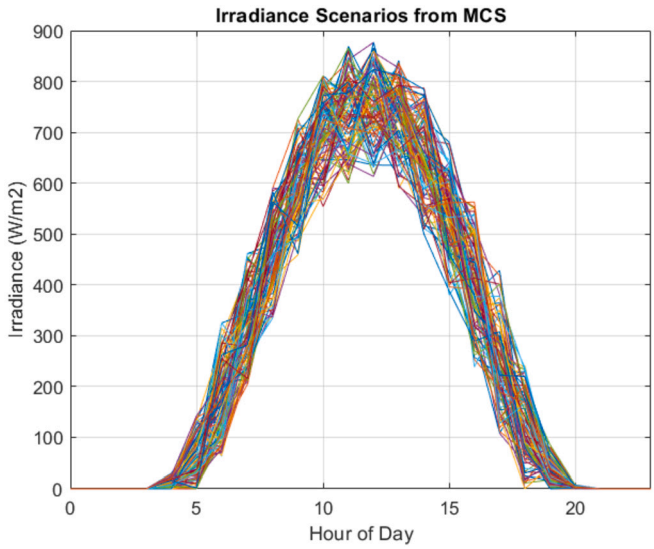


Fig. 12. Irradiance profile for 100 simulated scenarios using summer data.

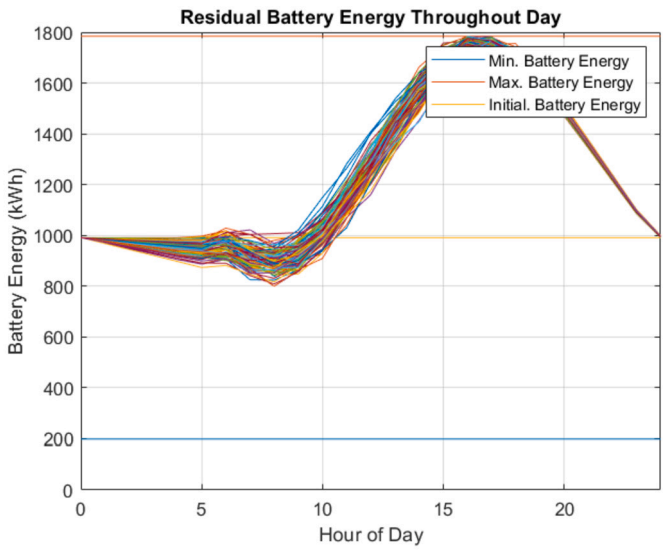


Fig. 13. Remaining battery energy during the day using summer data.

Table 10
Optimization results for PV and ESS sizing using summer data.

Optimization variable	Optimized value
PV size, kW	394.247
ESS (Battery capacity), kWh	2012.6
Overall Cost (Objective value), USD	2.4069 m

cost of the system be between that of the utilizing winter and summer data. Table 12 confirms this scenario. The results of these three different data scenarios are illustrated in Fig. 14.

To show the innovation in our developed model, we compare the developed model with other similar models in literature. Table 9 illustrates a comparison between key components of the developed model and other models within the research area.

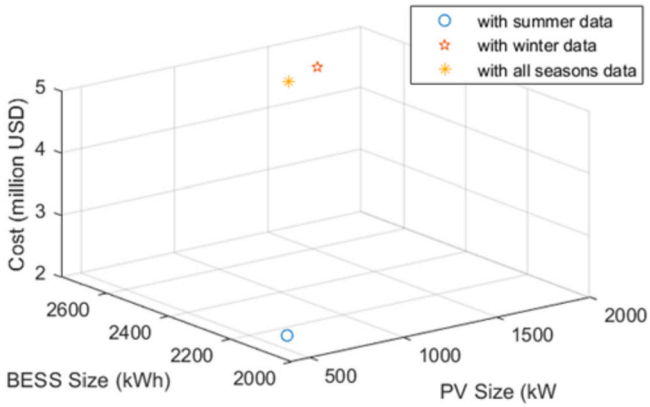


Fig. 14. Plot of reasons using different seasonal data.

Table 11
Optimization results for PV and ESS sizing using winter data.

Optimization variable	Optimized value
PV size, kW	1762.1
ESS (Battery capacity), kWh	2738.2
Overall Cost (Objective value), USD	4.5 m

Table 12
Optimization results for PV and ESS sizing for all seasons.

Optimization variable	Optimized value
PV size, kW	1610.8
ESS (Battery capacity), kWh	2741.9
Overall Cost (Objective value), USD	4.3527 m

5.2. Real world test case and parametric sweep analysis

As discussed in section 4.1, it is important to find the optimal balance between the battery size (cost) and the wireless charging stations built on the road. Furthermore, parameters such as the number of buses within the bus fleet have a significant effect on the choice of battery size as well as the complementary charging infrastructure needed on the road. In actual transportation systems, the number of buses that ply a route differs based on the type of day. More buses are used during peak periods and a relatively smaller number during off-peak periods.

To know the effects of different bus fleet numbers on the overall optimization–the battery size and power transmitter combination–a parametric sweep analysis is conducted. These results are presented in Figs. 15, 16, 17 and 18.

Fig. 15 plots the number of buses within the bus fleet against the optimal battery size and the required number of wireless charging stations. From Fig. 15, it is realized that when the number of buses is 7, the number of charging stations needed to be built is 0. The corresponding required battery capacity for this instance is approximately 18 kWh. However, as the number of buses in the fleet increases, it is more economical to balance smaller batteries with an increasing number of charging stations (and thus longer length of wireless power transmitters) as shown in Fig. 16. As the number of buses increases, the battery sizes tend to reduce but the extra costs in building the charging stations and the extra battery cost for the extra buses increases the overall cost of the systems. This is depicted in Fig. 17.

Furthermore, it is also of interest to know the separate costs of the batteries and the charging stations at each operating bus fleet number. When the number of buses within the fleet is 7, there is no charging station cost since no station will be built as shown in Fig. 18. As shown in Fig. 18, when the number of buses increases to 10, the drastic reduction in the battery size (from 18 kWh to 10.48 kWh) leads to a reduction in the overall battery costs but raises the cost of the charging stations since the required station to be built increases to one (total of 60 m). Next, when the number of buses in the fleet increases from 10 and 15 buses, the cost of the charging stations increases since the required station to be built increases from one (total of 60 m) to two (total of 87 m). During this period, the battery size required reduces from 10.48 kWh to 7.11 kWh. When the number of buses is further increased from 15 and 20, the most economical number of stations needed is two with a corresponding battery size of 7.11 kWh. However, as the total number of buses increased, the total cost of the batteries increased. The cost of the batteries then falls when the number of buses within the fleet is increased to 25 while the cost of the charging stations rises. This is because the optimal number of charging stations

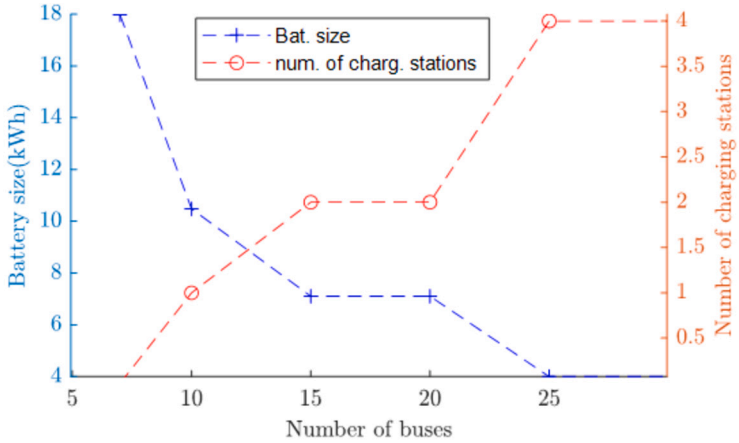


Fig. 15. Number of buses against battery size and number of charging stations.

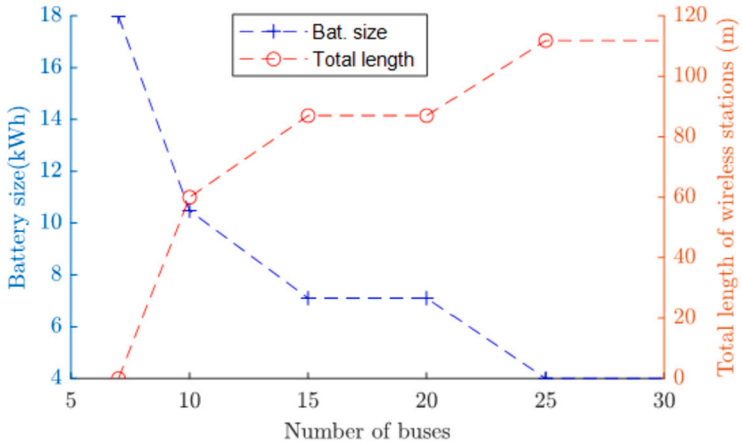


Fig. 16. Number of buses against battery size and total length of charging stations.

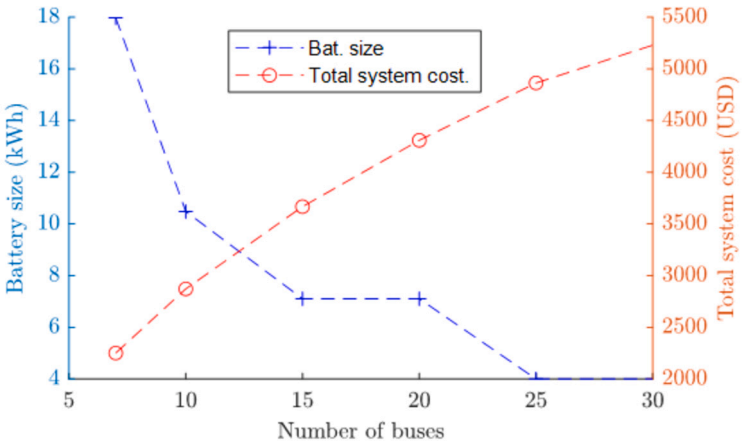


Fig. 17. Number of buses against battery size and total system cost.

rises to 4 (with a total length of 111.84 m) when the number of buses within the fleet is from 25 to 30. When the buses in the fleet are increased from 25 to 30, the total annual battery cost rises from USD 1788.4 to USD 2146.

The results of the parametric analysis give an idea of the various optimal operating points when the buses within a fleet that serves a particular route are varied. Table 13 and Table 14 present a summary of the results for real-world scenario and parameter analysis.

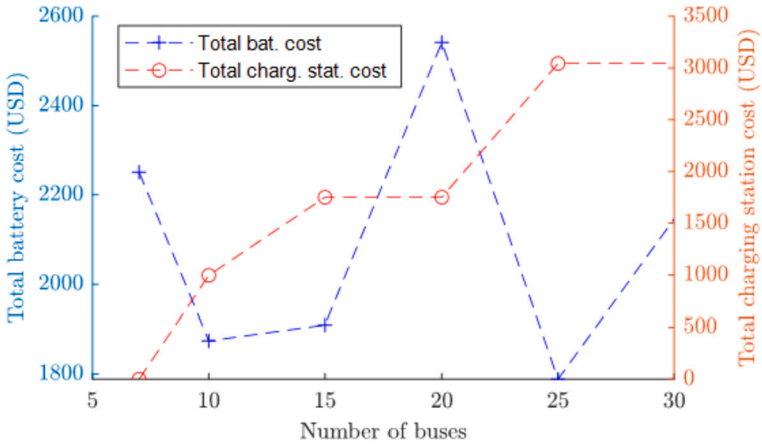


Fig. 18. Number of buses against total battery cost and total charging station cost.

Table 13
Summary of charging station placement locations and respective lengths.

Bus stop area	Number of buses in fleet					
	7	10	15	20	25	30
SAN schools						
NMC Royal						
National Aquarium						
Al Jazeera Club			32	32	30.321	30.321
Bint Said Mosque					30.125	30.125
Capital Park		60				
ADB bus station						
MBK street						
Al Jazeera Club			55	55		
National Aquarium					31.794	31.794
Al Tashreef					19.6	19.6
NMC Royal						
Total length (m)	0	60	87	87	111.84	111.84

Table 14
Optimization results for the objective function and bus battery capacity.

Number of buses in fleet	Bat. Cap. (kWh)	Object. Value (USD)
7	17.98	2250.8
10	10.48	2871.7
15	7.11	3667.56
20	7.11	4309.23
25	4	4863.89
30	4	5229.73

6. Conclusion

Wireless charging principles enable battery electric buses (BEBs) to charge wirelessly in three different ways—static, dynamic, and quasi-dynamic—using a wireless power transfer (WPT) technology other than charging cables. In quasi-dynamic charging (QWC) systems, the BEBs charge as they stop for brief periods in traffic or bus stops during their journeys. In this paper, first, a mathematical optimization model for the optimal siting of QWC stations in an electric-transport nexus is presented. The problem is formulated as a MINLP model. The goal of this optimization model is to optimally find the location and length of QWC stations (power transmitters) as well the optimal battery size for the bus fleet while focusing on minimizing the annual investment, operating cost of building wireless charging stations (power transmitters), and the cost of energy losses due to BEB charging. Two alternative optimization approaches are applied to find the solution to the problem and compared.

Secondly, an optimization model for sizing a PV and BESS system to meet the power demand of the BEBs charging wirelessly from the siting model is presented to supply the load requirements of the electric bus fleet. First, a bus schedule is made for the buses within the fleet to ascertain the number of buses that pass through a bus stop and thus, a power transmitter each hour. Then the

power demand of the bus fleet in the charging stations is computed. The load demand by the bus fleet per hour in several scenarios serves as an input to the developed optimization model.

The developed optimization model is implemented and solved in MATLAB using Casadi (with Bonmin solver) and Yalmip (with Fmincon solver) optimization toolboxes. From the developed optimization problem for the power transmitter siting, the Bonmin solver returned better results compared to the Fmincon. From the results and discussion, a 13 kWh battery capacity, is selected for the bus fleet and 5 power transmitters would be installed at 6 different bus stop areas with a total power transmitter length of 300 m and a total annual cost of USD 7330.32. In addition, the model recommends a suitable bus in an IEEE 13 bus system where the bus loads could be connected.

Next, the sizing model gives multiple results depending on the seasonal data used in the optimization. With three different seasonal (winter, summer and all seasons) data being used, three pairs of results are achieved for the PV and BESS sizing. These results are found to be a PV and BESS with a rating of 394.247 kW 2012.6 kWh, 1762.1 kW and 2738.2 kWh and 1610.8 kW and 2741.9 kWh for summer, winter and all-seasons (year-round) data respectively are needed to meet the energy requirements of the bus fleet throughout the day. These results also show how the type of seasonal data used for PV-ESS sizing could affect the sizing and cost of the system.

Furthermore, the developed charging station placement-battery sizing model was tested using a real-world example of a bus route between Khalifa City and Abu Dhabi Downtown in the Emirate of Abu Dhabi in the UAE with data from [66]. To know the effects of the different bus fleet numbers on the optimal wireless charging station location (including length of the power transmitters) and battery size combination, a parameter analysis is conducted. The results show that when the number of buses in the fleet is relatively low (set to 7 buses), it is more economical to use a larger battery size with no power transmitters installed along the route. With an increase in number of the buses in the fleet, the results show that the more economical solution is a combination of a relatively smaller battery size and an increasing number (and length) of power transmitters.

7. Future work

The developed model focuses on BEBs as the model assumptions prioritize bus systems where the exact timings and routes are clearly defined. Major uncertainties exist in the case of general and individual EV users. These uncertainties could arise from driving patterns, weather conditions, and locations of charging stations. The uncertainties in behavioral and driving patterns of general drivers could be characterized towards predicting the locations as well as the length of the power transmitters needed. In addition, a real-time monitoring approach could be developed to ascertain the level of charge of the EV batteries and recommend the closest charging stations based on their real-time locations. Furthermore, an EV load growth model could give an idea of the extent of the load increase and plans could be made for the extension of the wireless charging stations as well as the necessary PV and BESS rating required to meet future load requirements.

CRediT authorship contribution statement

Prince Aduama: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ameena S. Al-Sumaiti:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Khalifa H. Al-Hosani:** Writing – review & editing, Supervision, Funding acquisition. **Ahmed R. El-Shamy:** Writing – review & editing, Visualization, Software, Investigation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All data used has been included/cited in the manuscript.

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